Day 09 | Tuesday

November 10, 2026

09

Repeated pieces and equal groups

12:45 PM–2:00 PM • Math • 75 min

Fraction operations: check and retry

2:15 PM–3:30 PM • Math • 75 min

Week 3 • 2 scheduled blocks • 150 teaching minutes

Estimate. Show the method. Label units. Check the result.

Today’s exact schedule

12:45 PM–2:00 PM • 75 min

Repeated pieces and equal groups

12:45 PM–12:50 PM	**Start**	5
12:50 PM–12:55 PM	**Recall**	5
12:55 PM–1:10 PM	**Teach**	15
1:10 PM–1:25 PM	**Together**	15
1:25 PM–1:45 PM	**Practise**	20
1:45 PM–1:55 PM	**Check**	10
1:55 PM–2:00 PM	**Wrap up**	5

Math • Math Week 3, Lesson 3

2:15 PM–3:30 PM • 75 min

Fraction operations: check and retry

2:15 PM–2:20 PM	**Start**	5
2:20 PM–2:25 PM	**Recall**	5
2:25 PM–2:40 PM	**Teach**	15
2:40 PM–2:55 PM	**Together**	15
2:55 PM–3:15 PM	**Practise**	20
3:15 PM–3:25 PM	**Check**	10
3:25 PM–3:30 PM	**Wrap up**	5

Math • Math Week 3, Lesson 4

Break: 2:00 PM–2:15 PM • 15 minutes • excluded from teaching time

Repeated pieces and equal groups

Today’s goal: Use multiplication and division to find repeated fractional lengths and centre lines.

Name one use for today’s skill.

Textbook Chapter 4 pp.67–70

Exercise Book 4A Q16–17 (p.26); 4B Q4–5 (p.27); 4C Q5 (p.30)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 4 pp.67–70

Recall and reconnect

Explain how repeated addition connects to multiplication and how dividing by two finds a centre.

Try first without displayed answers.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q4–5 (p.27); 4C Q5 (p.30)

Textbook Chapter 4 pp.67–70

Multiply to find repeated fractional lengths

1. Convert a mixed number to an improper fraction when useful.
2. Multiply the numerators and multiply the denominators.
3. Simplify and convert back to a mixed number when needed.
4. A repeated-strip drawing checks whether the total fits the number of pieces.

Textbook Chapter 4 pp.67–70

Explain what division counts or measures

1. Dividing a length by a piece length gives a count of pieces.

2. Dividing a length by a number of equal parts gives a length per part.

3. To divide by a fraction, multiply by the reciprocal of the divisor.

4. Check a centre line by doubling its distance from the starting edge.

Textbook Chapter 4 pp.67–70

Ask what the result measures or counts

3 × 2 3/8 in = 3 × 19/8 = 57/8 = 7 1/8 in

3/4 in ÷ 1/8 in = 3/4 × 8/1 = 6 pieces

The reciprocal changes division to multiplication.
Check the second result: six eighth-inch pieces total 3/4 in.
These classroom examples ignore kerf.

Together: 4A Q16(a), Q17(a), then 4B Q5 with a centre-line sketch. Repeat a labelled Shape Canvas rectangle and compare totals.

Textbook | page 68

Example 4.12 How many pieces?

How many spacers $1\frac{3}{4}$ in long can be cut from a 24 in bar? Ignore the saw cuts. How long is the offcut?

SOLUTION

1. $24 \div 1\frac{3}{4} = \frac{24}{1} \times \frac{4}{7} = \frac{96}{7} = 13\frac{5}{7}$.
2. Only whole spacers count: 13 spacers.
3. 13 spacers use $13 \times 1\frac{3}{4} = 22\frac{3}{4}$ in. The offcut is $24 - 22\frac{3}{4} = 1\frac{1}{4}$ in.

Textbook p.68 · Example 4.12: divide fractional lengths and count complete pieces.

Textbook | page 68

CALCULATOR

TI-30Xa

Spacers

24 ÷ 1 a b/c 3 a b/c 4 =

Display: 13⌟5⌟7

Offcut

24 − 13 × 1 a b/c 3 a b/c 4 =

Display: 1⌟1⌟4

Sharp EL-531XT

Spacers

24 ÷ 1 a b/c 3 a b/c 4 =

Display: 13⌜5⌜7

Offcut

24 − 13 × 1 a b/c 3 a b/c 4 =

Display: 1⌜1⌜4

The first display reads $13\frac{5}{7}$. Only the whole number of spacers counts.

Textbook p.68 · Exact fraction calculator entry for the pieces and offcut.

Spot the mistake. Explain the correction.

For 3/4 in ÷ 1/8 in, should the answer be labelled inches?

Discuss the method before checking the answer.

The quotient counts eighth-inch pieces: 6 pieces. Check that 6 × 1/8 in restores 3/4 in.

Textbook Chapter 4 pp.67–70

Work together

Work 4A questions 16(a) and 17(a), then 4B question 5 with a centre-line sketch.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q4–5 (p.27); 4C Q5 (p.30)

Textbook Chapter 4 pp.67–70

Guided practice | 4A Q16a

Multiply. (a) $\frac{5}{8} \times 12$

Work with a partner. Take turns explaining the next step.

Textbook Chapter 4 pp.67–70

Compare your method | 4A Q16a

Multiply. (a) $\frac{5}{8} \times 12$

CHECK TOGETHER

(a) 7 1/2

Textbook Chapter 4 pp.67–70

Your scheduled practice

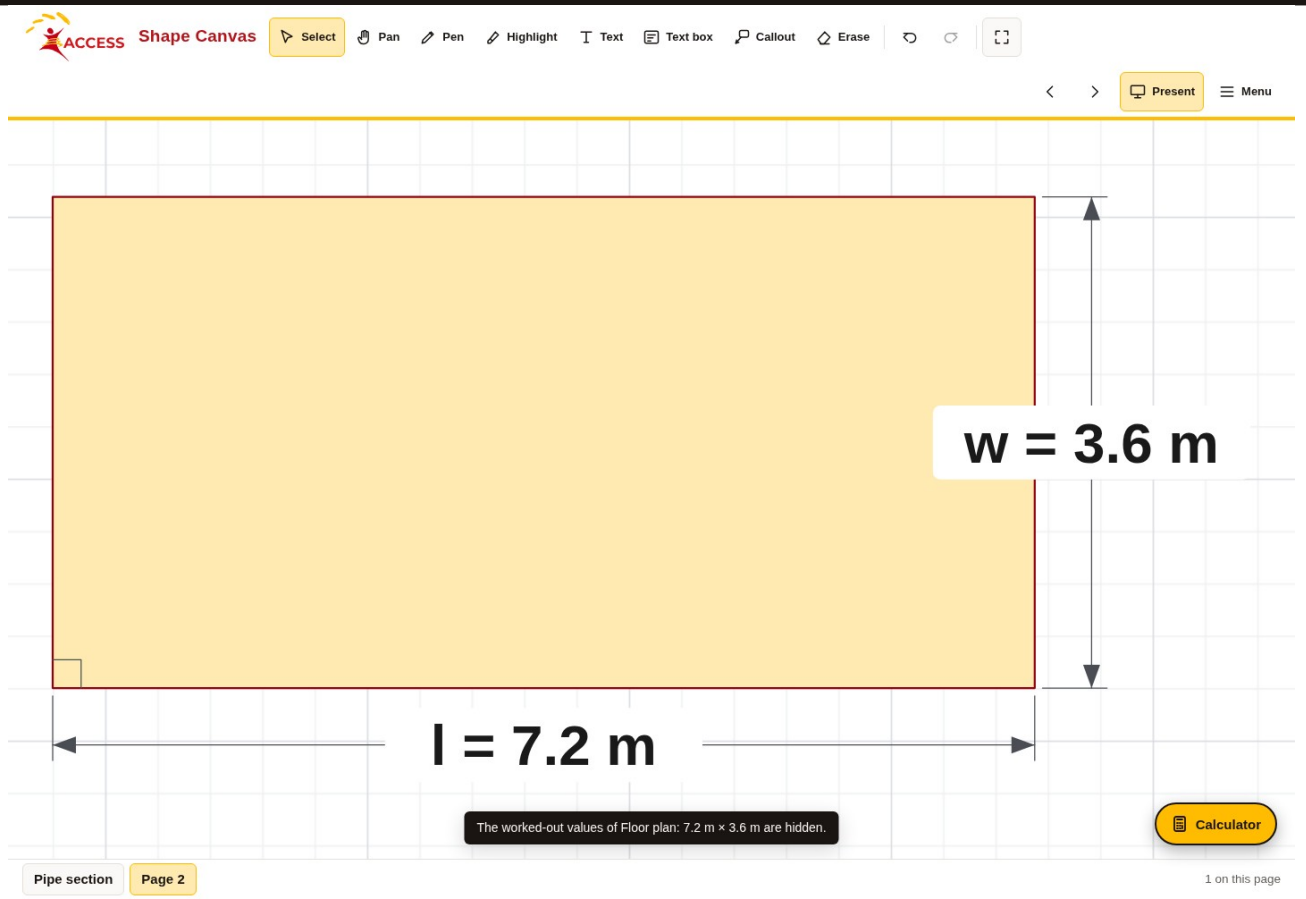
Complete remaining parts of 4A questions 16–17, 4B question 4 and 4C question 5. Explain what the division result counts or measures.

Show working and units; estimate, then check.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q4–5 (p.27); 4C Q5 (p.30)

Textbook Chapter 4 pp.67–70

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 4 pp.67–70

Practise in Numeracy Hub and Shape Canvas

1. Supported: Repeat a labelled Shape Canvas rectangle to show multiplication as equal pieces.

2. Independent practice: Write exact fraction calculations for the assigned 4A, 4B and 4C items.

3. Independent checking: Use Numeracy Hub Calculator; label each answer as a length or a piece count.

Textbook Chapter 4 pp.67–70

Practise from the Exercise Book | 4A Q16b

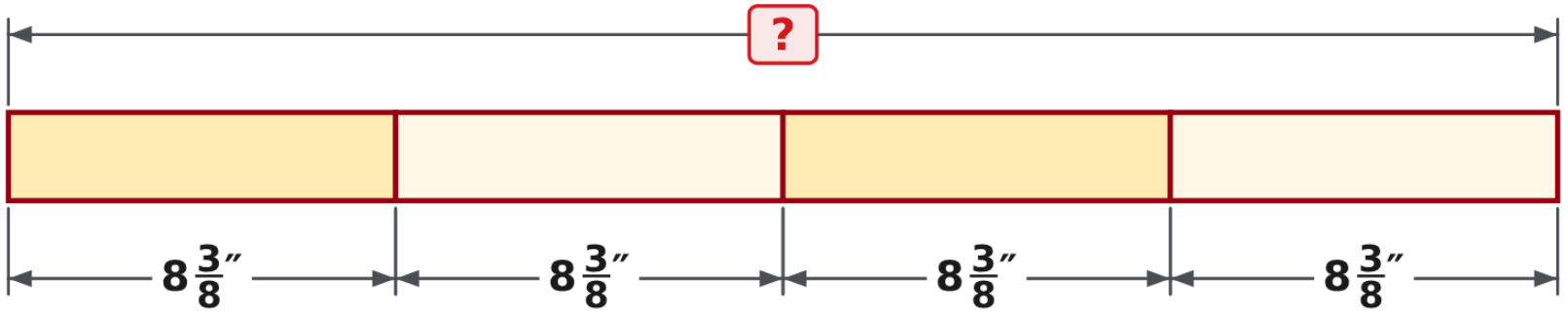
Multiply. (b) $2\frac{1}{2} \times 1\frac{3}{4}$

Write the operation, working and units before checking.

Textbook Chapter 4 pp.67–70

Practise from the Exercise Book | 4C Q5

Change one value: four equal pieces of $8\frac{3}{8}$ in make up the bar. Find the overall length. If each piece is made $\frac{1}{16}$ in longer, what is the new overall length?



Write the operation, working and units before checking.

Textbook Chapter 4 pp.67–70

Check independently

Complete 4A question 17(b) and 4B question 5.

Answer independently and show the method.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q4–5 (p.27); 4C Q5 (p.30)

Textbook Chapter 4 pp.67–70

Independent check | 4A Q17b

Divide. (b) $15\frac{3}{4} \div 3$

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.67–70

Independent check | 4B Q5

A plate is $15\frac{5}{8}$ in wide. How far from the edge is the centre line?

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.67–70

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 4A Q16–17 (p.26); 4B Q4–5 (p.27); 4C Q5 (p.30)

Textbook Chapter 4 pp.67–70

Break

2:00 PM–2:15 PM

Return ready for: Fraction operations: check and retry

This break is excluded from teaching minutes.

Fraction operations: check and retry

Today’s goal: Choose the correct fraction operation, show the working and correct a documented error.

Name one use for today’s skill.

Textbook Chapter 4 pp.64–70

Exercise Book 4A Q14–17 (p.26); 4C Q1–3 (p.29)

Keep your notebook, textbook, exercise book and chosen calculator ready.

Textbook Chapter 4 pp.64–70

Recall and reconnect

Present four short situations: total, remainder, repeated pieces and equal sharing. Name the operation.

Try first without displayed answers.

Resources: Exercise Book 4A Q14–17 (p.26); 4C Q1–3 (p.29)

Textbook Chapter 4 pp.64–70

Choose the method from the operation

1. Addition and subtraction need common denominators.

2. Multiplication multiplies numerators and denominators.

3. Division multiplies by the reciprocal of the divisor.

4. Choose the operation from the situation before applying a procedure.

Textbook Chapter 4 pp.64–70

Make the correction explain the error

1. Keep the original working so the incorrect step can be identified.
2. State whether the error is in the operation, denominator, regrouping or reciprocal.
3. Write a corrected method and explain why it preserves the quantities.
4. Use a fresh partner-created value to check the same skill without copying.

Textbook Chapter 4 pp.64–70

Explain why the first method failed

Incorrect: 1/2 + 1/4 = 2/6.
The denominators were added even though the pieces differ in size.

Correct: 2/4 + 1/4 = 3/4

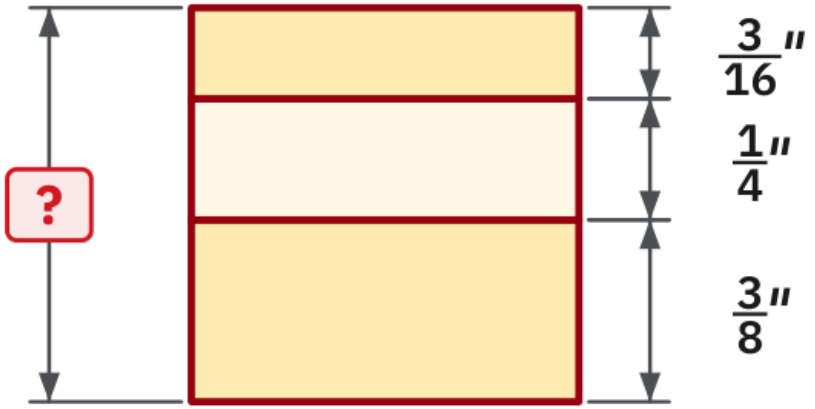
Contrast multiplication: 1/2 × 1/4 = 1/8.
Here the question asks for half of a quarter, not a combined length.

Together: explain one method from 4A Q14–17 without copying a partner. Instructor models the most common error from Monday's work.

Textbook | page 64

Example 4.8 Stacking plates

Three plates are stacked. They are $\frac{3}{8}$ in, $\frac{1}{4}$ in and $\frac{3}{16}$ in thick. How thick is the stack?



SOLUTION

1. Change to sixteenths: $\frac{6}{16}$, $\frac{4}{16}$ and $\frac{3}{16}$.
2. Add the numerators: $6 + 4 + 3 = 13$.
3. The total is $\frac{13}{16}$. 13 is odd, so it is in lowest terms.

Textbook p.64 · Example 4.8: use equal-sized fractions to add a stack.

Textbook | page 64

CALCULATOR

TI-30Xa

3 | a b/c | 8 | + | 1 | a b/c | 4 | + | 3

a b/c | 16 | =

Display: 13⌟16

Sharp EL-531XT

3 | a b/c | 8 | + | 1 | a b/c | 4 | + | 3

a b/c | 16 | =

Display: 13⌐16

The TI-30Xa shows 13⌟16 and the Sharp shows 13⌐16. Both mean $\frac{13}{16}$.

Textbook p.64 · Check the written fraction method on the chosen calculator.

Spot the mistake. Explain the correction.

Must you find a common denominator before every fraction multiplication?

Discuss the method before checking the answer.

No. Common denominators support addition and subtraction. Multiply numerators and denominators directly.

Textbook Chapter 4 pp.64–70

Work together

Pairs explain a method from 4A questions 14–17 without copying the partner’s calculation.

Work in pairs; alternate solver and checker.

Resources: Exercise Book 4A Q14–17 (p.26); 4C Q1–3 (p.29)

Textbook Chapter 4 pp.64–70

Guided practice | 4A Q14

Add. (a) $\frac{5}{16} + \frac{3}{4}$ (b) $7\frac{3}{8} + 2\frac{15}{16}$

Work with a partner. Take turns explaining the next step.

Textbook Chapter 4 pp.64–70

Compare your method | 4A Q14

Add. (a) $\frac{5}{16} + \frac{3}{4}$ (b) $7\frac{3}{8} + 2\frac{15}{16}$

CHECK TOGETHER

(a) 1 1/16 (b) 10 5/16

Textbook Chapter 4 pp.64–70

Your scheduled practice

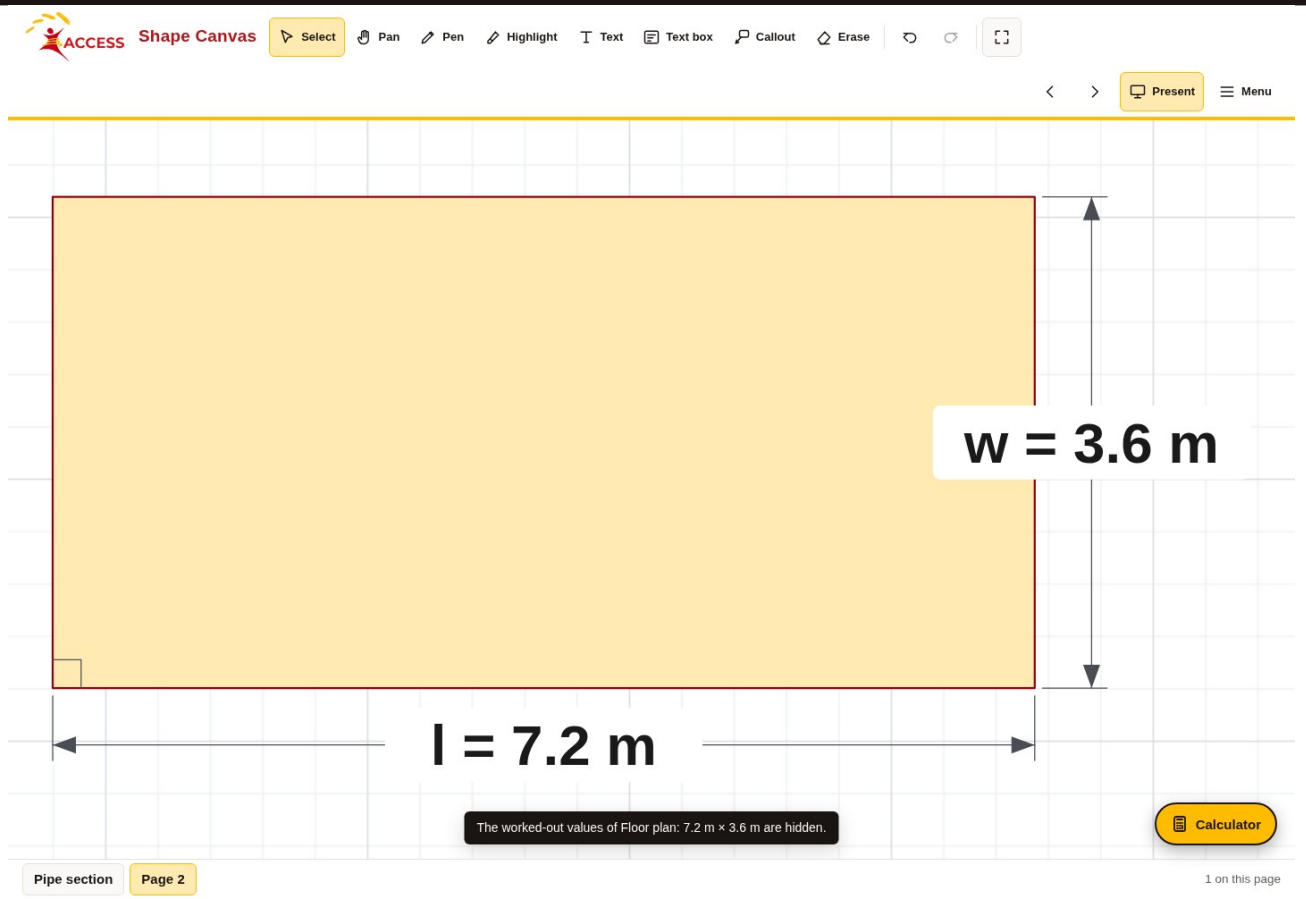
Complete one part of each question 14–17 independently, then 4C questions 1–3. Mark, explain and retry one error with a fresh partner-created value.

Show working and units; estimate, then check.

Resources: Exercise Book 4A Q14–17 (p.26); 4C Q1–3 (p.29)

Textbook Chapter 4 pp.64–70

Open Shape Canvas



Navigation example. Use the values and units in the assigned task.

Textbook Chapter 4 pp.64–70

Practise in Shape Canvas

1. Independent first attempt: Complete one part of 4A Q14–17 and the assigned 4C drawings on paper.

2. Checking after completion: Use the chosen calculator and explain one difference from your result.

3. Independent retry: Rebuild one missing-length sketch in Shape Canvas and solve a fresh missed type.

Textbook Chapter 4 pp.64–70

Practise from the Exercise Book | 4A Q14a

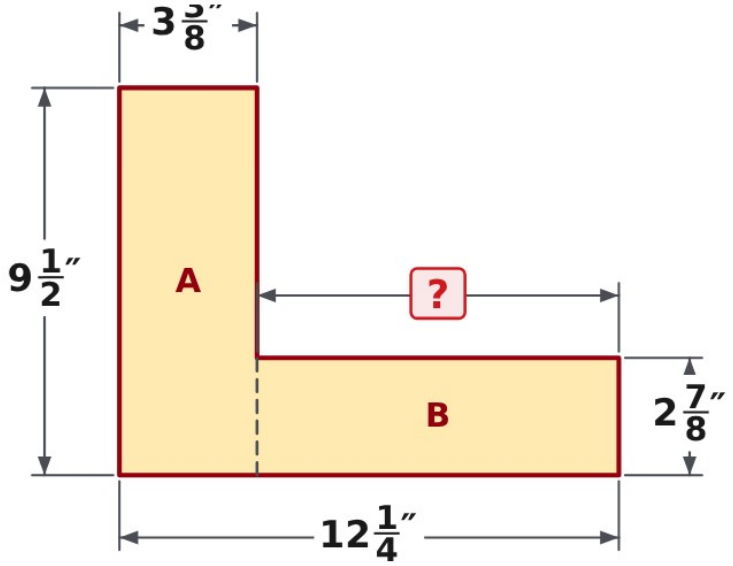
Add. (a) $\frac{5}{16} + \frac{3}{4}$

Write the operation, working and units before checking.

Textbook Chapter 4 pp.64–70

Practise from the Exercise Book | 4C Q1

Find the hidden width.



Write the operation, working and units before checking.

Textbook Chapter 4 pp.64–70

Check independently

Redo 4A questions 14(a),15(a),16(b),17(b). Give simplest forms.

Answer independently and show the method.

Resources: Exercise Book 4A Q14–17 (p.26); 4C Q1–3 (p.29)

Textbook Chapter 4 pp.64–70

Independent check | 4A Q14a

Add. (a) $\frac{5}{16} + \frac{3}{4}$

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.64–70

Independent check | 4A Q15a

Subtract. (a) $5 - 1\frac{11}{16}$

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.64–70

Independent check | 4A Q16b

Multiply. (b) $2\frac{1}{2} \times 1\frac{3}{4}$

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.64–70

Independent check | 4A Q17b

Divide. (b) $15\frac{3}{4} \div 3$

Answer independently. Keep worked solutions closed.

Textbook Chapter 4 pp.64–70

Wrap up and choose the next step

Share one learning point and one skill to practise next.

Record a takeaway and next action.

Resources: Exercise Book 4A Q14–17 (p.26); 4C Q1–3 (p.29)

Textbook Chapter 4 pp.64–70